Dynamic Cheap Talk with Endogenous States*

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Abstract

I study cheap talk between a long-lived Sender and a sequence of short-lived Receivers. The distribution of the Sender's private information depends on one of two publicly observed states, and each Receiver's action affects which state occurs next. Stage payoffs are identical, so in a static setting the Sender would truthfully reveal everything a Receiver needs to know. Because only the Sender internalises the effect of today's action on future states, the Sender becomes endogenously biased towards the action that makes the more valuable state more likely. Communication breaks down only when actions affect future states strongly enough and the bias works against the Receiver's prior. Communication in one state affects its credibility in the other, the Sender attains the first-best payoff if and only if communication is possible in both states, and full revelation requires that actions not affect the state.

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1 Introduction

In many situations, a decision-maker lacks the information needed to make a decision, and seeks advice from someone who is better informed: investors consult financial advisors, buyers listen to sales representatives, and voters listen to politicians. Since Crawford and Sobel (1982), a large literature has studied how much information such an expert can credibly convey when her interests differ from the decision-maker's. In most of this literature, the conflict of interest is a primitive: the expert is biased in the one-shot interaction, and the larger her bias, the less she can communicate. I focus on a conflict that arises even when the expert and the decision-maker agree on what is best today. This is because the decision-maker's action affects the environment in which future decisions are made, and only the expert cares about the future.

Consider a government that advises citizens on whether to go out during an epidemic. The government knows more than the citizens about current risks, and on any given day, the two agree on whether going out is worthwhile. But going out also raises transmission, and hence the risk of a surge. Each citizen, deciding only for himself today, ignores this effect, whereas the government cares about the course of the epidemic. Whether its advice can then be trusted is a natural question: Dodsworth (2021), for instance, argues that the UK government "weaponised fear" during the Covid-19 pandemic to induce people to stay at home.

This paper studies how an advisor communicates when she cares about the future, and when her advice is credible. I consider a dynamic cheap-talk game between a long-lived Sender and a sequence of short-lived Receivers. In each period, the Sender privately observes the realisation of a random variable whose distribution changes over time: I call the realisation her *type*, and the distribution the *state*. For simplicity, there are two states, low and high, which both players observe. The Sender sends a costless message about her type, and the Receiver then chooses between a low and a high action, which affects the probability that the next state is high. Since her message influences the Receiver's action, the Sender has some control over the distribution of her future types, and cheap talk is her only instrument for exercising it. The Sender and the Receivers have identical stage payoffs, and the Sender maximises the discounted sum of the Receivers' payoffs. In a static setting, she would therefore tell the Receiver everything he needs to know. The only

source of conflict is that the Receiver's action imposes an externality on future Receivers, through its effect on future states, which he ignores and the Sender internalises. I ask when informative communication is possible, what the players' equilibrium payoffs are, and when the Sender attains her first-best payoff.

I first show that, although the Sender is unbiased in the stage game, she becomes endogenously biased in the dynamic game: the externality creates her bias and determines its direction. In a two-period version of the model, the Sender prefers the high action if and only if her type exceeds a cutoff, which differs from the Receiver's by an amount proportional to the effect of the high action on the probability of reaching the high state. She is therefore biased towards the action that makes the high state more likely. Her recommendations are credible only if this bias does not work too strongly against the Receiver's prior (Proposition 1): a bias in the direction of the prior never undermines communication, while a strong enough bias against it leads the Receiver to ignore her messages (Corollary 1).

In the infinite-horizon model, the value of each state to the Sender is itself an equilibrium object, since it depends on how well she can communicate in each state. I focus on Markov perfect equilibria (MPE). In every MPE, the Sender uses a cutoff in each state in which she communicates, and her payoff is a weighted average of the Receivers' expected payoffs in the two states, with weights given by the discounted frequencies with which each state is visited (Lemma 2). As she becomes patient, the weights converge to the invariant distribution of the state, which, since transitions depend on the Receivers' actions, depends on how she communicates. Communication therefore affects the Sender's payoff through two channels: the quality of the Receivers' decisions in each state, and the share of time spent in each state. Her bias reflects the trade-off between the two. She distorts today's decision in proportion to the gap between the Receivers' payoffs in the two states, multiplied by the effect of today's action on the expected discounted time spent in the high state. If she could choose the action herself, her optimal cutoffs would be the unique solution to a system of two equations, and the high state would be more valuable to her, so her bias has the same direction as in the two-period model (Proposition 2).

There are four types of MPE: the Sender communicates in both states, in only one of them, or

in neither (Proposition 3). She can communicate in a state if and only if the Receiver is better off following her recommendations than ignoring them, that is, if and only if her bias does not work too strongly against his prior there. A bias towards the high action thus threatens communication in the low state, where the Receiver is inclined to take the low action, and a bias towards the low action threatens it in the high state. Broadly, the bias is larger when the Sender is more patient, when the Receiver's action has a larger effect on transitions, and when the gap between the Receivers' expected payoffs in the high and the low state is larger. Since this gap depends on communication in both states, the existence conditions are joint across states: whether the Sender communicates in one state changes the Receivers' expected payoff there, and hence her bias in the other. For some parameters, for instance, she can communicate in the high state only if she also communicates in the low state.

The types of MPE differ in efficiency. Since the Sender maximises the discounted sum of the Receivers' payoffs, her first-best is the efficient outcome, and it involves a bias whenever actions affect transitions: the current Receiver loses relative to full revelation, but future Receivers gain more. The Sender attains her first-best payoff in an MPE if and only if she can communicate in both states (Corollary 3), so an MPE in which she communicates in both states is efficient, while, except in boundary cases, the others are not. Full revelation, which maximises each Receiver's own payoff, is an MPE only when actions do not affect transitions (Corollary 4): only then does the Sender have no control over future states, and hence no reason to depart from the Receiver's preferred action. Finally, since the effect of today's action on the state persists over many periods, the externality can undermine communication in the infinite-horizon model at values at which it would not in the two-period model.

These results have two implications. First, they apply directly to the epidemic. The government is the Sender, and the citizens, each of whom decides only for himself today, are the Receivers. The state is the phase of the epidemic, contained or surging, which both observe, and the type is the share of the population that is currently healthy, which only the government observes through testing. Going out is the high action: citizens prefer it when enough of the population is healthy, but it raises transmission and makes a surge more likely. The government is therefore biased towards

recommending that citizens stay at home, and in effect overstates the share of the population that is infected. An advisor who exaggerates risks need not have different preferences from the people she advises: caring about the future is enough. Moreover, as long as citizens follow its advice in both phases, the government's exaggeration is efficient: it sacrifices some of today's welfare to keep the epidemic contained for longer. The same concern also limits the credibility of the government's advice. While the epidemic is contained, citizens are inclined to go out, so a recommendation to stay at home works against their prior, and it can be ignored precisely when going out is most likely to trigger a surge. Communication in one phase can, however, sustain communication in the other: for some parameters, a government that communicates during a surge keeps its advice credible once the epidemic is contained, whereas one that does not would be ignored.

Second, in contrast to the literature, the Sender's bias is not a primitive but is determined jointly with communication in both states. In dynamic cheap-talk models with exogenous states, the long-run distribution of the state does not depend on communication. Here it does, so the Sender's long-run payoff depends on the type of equilibrium that is played.

Section 1.1 discusses the related literature. Section 2 presents the model, Section 3 analyses a two-period version, Section 4 studies the infinite-horizon model, and Section 5 concludes. Proofs not given in the text are in the appendices.

1.1 Related Literature

This paper contributes to the literature on cheap talk that started with Crawford and Sobel (1982) and Green and Stokey (2007). The canonical model has been extended in many directions, including longer communication protocols, reputational concerns, multiple senders and receivers, and costly information acquisition; Sobel (2015) provides an overview. In these models, the Sender and the Receiver have an exogenous conflict of interest in the one-shot game, and this conflict limits how much information can be transmitted. Here, the stage-game preferences are identical, and the conflict arises from the dynamics: the Sender's bias depends on the values of the two states, which in turn depend on equilibrium communication. If the high action makes the high state more likely in one state and less likely in the other, the direction of the bias differs across states; Gordon (2010)

studies cheap-talk games in which the direction of the Sender's bias differs across her types.

Dynamic cheap-talk models differ in how the Sender's information evolves. In some, the Sender's type is fixed. Anderlini, Gerardi, and Lagunoff (2012) study an infinite sequence of decision-makers, each of whom both receives and sends messages about a fixed state, and show that full learning is impossible under both public and private communication. Golosov et al. (2014) study a finite-horizon version of Crawford and Sobel (1982) with two long-lived players, and show that full revelation can be achieved in finite time, because future communication can be made contingent on the Receiver's past actions. In others, types are drawn independently over time. Avery and Meyer (2012) study the reputational incentives of a possibly biased evaluator who advises on binary decisions. As in their static model, in each state of my model, an informative equilibrium, if it exists, is one in which the Sender uses a cutoff. Best and Quigley (2024) study a long-lived Sender and short-lived Receivers who observe a public record of feedback on the Sender's past messages, and ask when such a record can substitute for commitment. Finally, in Renault, Solan, and Vieille (2013), the Sender's information follows an exogenous Markov chain and both players are long-lived. They show that, as the players become patient, the set of equilibrium payoffs depends on the chain only through its invariant distribution. In all these papers, the process that governs the Sender's information is exogenous. Here, the Receiver's action affects it,[1] and this is the only source of conflict. As a result, the invariant distribution of the state, and hence the Sender's long-run payoff, depends on how she communicates.

Finally, from a modelling perspective, the game is a stochastic game (Shapley, 1953; Solan & Vieille, 2015) in which one player's action affects the transition probabilities, as in Renault (2012). There, however, the controlling player is also informed, whereas here the Receivers control the transitions and hold no private information. In repeated games with incomplete information, the informed player's private information is either fixed (Aumann & Maschler, 1995) or follows an exogenous Markov chain (Renault, 2006; Escobar & Toikka, 2013); in both cases it is persistent, so revealing it today affects how it can be used in the future. Here, the Sender's type is drawn afresh each period given the publicly observed state, so the dynamics operate only through the effect of

[1] In Kalantri (2026), I study information provision with commitment in an environment in which the Receivers' actions also impose an intertemporal externality.

the Receivers' actions on the state.

2 Model

I study a dynamic cheap-talk model with an infinite horizon. Time, indexed by $t \in \{0, 1, 2, \ldots\}$, is discrete.

Players. A long-lived, informed Sender (she) interacts with a sequence of short-lived, uninformed Receivers (he), one in each period. The period-t Receiver chooses an action $a_t \in A \coloneqq \{0, 1\}$, which I call the *low* and the *high* action.

States and types. In each period, the environment is in one of two states, $\omega_t \in \Omega \coloneqq \{L, H\}$ (low and high), which both players observe. The state determines the distribution of the Sender's type $x_t \in [0, 1]$, which is her private information. When $\omega_t = i$, x_t is drawn from a continuously differentiable distribution function F_i with strictly positive density f_i. Conditional on the sequence of states, types are drawn independently across periods. The initial state ω_0 is given and common knowledge. I write $\mathbb{E}_i$ for expectations under F_i, and $q_i \coloneqq \mathbb{E}_i[x]$ for the prior mean in state i.

Payoffs. When the Sender's type is x_t, the period-t Receiver's payoff from action a_t is:

$$U_{R,t} \coloneqq u(a_t, x_t), \qquad u(a, x) \coloneqq 2ax - a^2 = \begin{cases} 0, & a = 0, \\ 2x - 1, & a = 1. \end{cases} \tag{2.1}$$

The high action is therefore strictly better than the low action if and only if $x > 1/2$.[2] Being short-lived, each Receiver maximises his expected payoff in the current period given his information, and takes the low action when indifferent.

The Sender maximises the discounted average of the Receivers' payoffs. Given the initial state ω_0, her payoff is:

$$U_S \coloneqq \mathbb{E}\Big[(1-\delta)\sum_{t=0}^{\infty} \delta^t U_{R,t} \Big| \omega_0\Big] = \mathbb{E}\Big[(1-\delta)\sum_{t=0}^{\infty} \delta^t (2a_t x_t - a_t^2) \Big| \omega_0\Big], \tag{2.2}$$

[2] Equivalently, $u(a, x) = x^2 - (a - x)^2$: a quadratic loss, normalised so that the low action yields zero. Only the payoff difference $u(1, x) - u(0, x) = 2x - 1$ matters for the analysis.

where $\delta \in (0,1)$ is her discount factor. The Sender can thus be thought of as a benevolent advisor or planner: she and the Receivers have identical preferences in the stage game, so any conflict of interest between them is purely intertemporal.

I impose two assumptions on the type distributions. Under full information, the Receiver takes the high action if and only if $x > 1/2$, so his expected payoff in state i is:

$$v_i := \int_{1/2}^{1} (2x-1) f_i(x)\,\mathrm{d}x, \qquad i \in \Omega. \tag{2.3}$$

Assumption 1 (High state)**.** $v_H > v_L$.

Assumption 2 (Prior actions)**.** $q_L \leq 1/2 < q_H$.

Assumptions 1 and 2 compare the two states at the two extremes of information. Under full information, the Receiver's expected payoff in state i is v_i, and Assumption 1 says that it is higher in the high state. It holds, for instance, if F_H first-order stochastically dominates F_L, with $F_L(x) > F_H(x)$ for some $x \in (1/2, 1)$.[3] Under no information, the Receiver takes the action that is optimal under the prior. Assumption 2 says that this is the low action in state L and the high action in state H, so his expected payoff is 0 in state L and $2q_H - 1 > 0$ in state H: the high state is again better.[4]

State transitions. The state evolves endogenously: the distribution of ω_{t+1} depends on the current state and on the Receiver's current action. For $i \in \Omega$, the transition probabilities are:

$\omega_t = i$	$\omega_{t+1} = L$	$\omega_{t+1} = H$
$a_t = 0$	z_i	$1 - z_i$
$a_t = 1$	$1 - o_i$	o_i

(2.4)

[3] Since $v_i = \mathbb{E}_i[\max\{2x-1, 0\}]$ and $x \mapsto \max\{2x-1, 0\}$ is increasing, first-order stochastic dominance gives $v_H \geq v_L$ (Shaked & Shanthikumar, 2007, equation (1.A.7)). The inequality is strict because $v_i = 2\int_{1/2}^{1} \big(1 - F_i(x)\big)\,\mathrm{d}x$ and, by continuity, $F_L > F_H$ on an interval in $(1/2, 1)$.

[4] The assumption also determines which of the Receiver's obedience constraints can bind in each state; see Section 3.2.

where $z_i, o_i \in [0,1]$ are common knowledge.[5] The key quantity is the effect of the high action on the probability of reaching the high state:

$$\varepsilon_i \coloneqq z_i + o_i - 1 = \mathbb{P}(\omega_{t+1} = H \mid \omega_t = i, a_t = 1) - \mathbb{P}(\omega_{t+1} = H \mid \omega_t = i, a_t = 0) \in [-1, 1]. \tag{2.5}$$

Since the Receiver ignores the future, ε_i measures the *externality* his action in state i imposes on future Receivers. Several special cases are of interest.

(a) *Exogenous transitions* ($\varepsilon_L = \varepsilon_H = 0$): the state follows a Markov chain that does not depend on actions. This includes absorbing states: L is absorbing if $z_L = 1$ and $o_L = 0$, and H is absorbing if $z_H = 0$ and $o_H = 1$. If both states are absorbing, the state is fixed throughout the game.

(b) *Positive association* ($\varepsilon_L, \varepsilon_H > 0$): in both states, the high action raises the probability of the high state next period. *Negative association* ($\varepsilon_L, \varepsilon_H < 0$): in both states, it lowers that probability.

(c) *Mixed association* ($\varepsilon_L \varepsilon_H < 0$): the high action helps the future in one state and harms it in the other.

Timing. Each period proceeds as follows.

1. Both players observe the state ω_t. The Sender privately observes her type x_t.
2. The Sender sends a costless, unverifiable message $m_t \in M$, where $|M| \geq |A|$, so that the size of the message space never limits communication.
3. The Receiver observes m_t, updates his beliefs about x_t, and chooses a_t.
4. The stage payoff $u(a_t, x_t)$ is realised, and ω_{t+1} is drawn according to the transition probabilities in (2.4).

[5] Here, z_i is the probability that the next state is L after action 0 (*zero*), and o_i is the probability that the next state is H after action 1 (*one*).

Each Receiver observes the current state and the history of past states, messages and actions, but not past types or realised payoffs. Under the Markov strategies studied below, this history plays no role.

Strategies and equilibrium. I study Markov perfect equilibria (MPE): strategies depend on the history only through the current state and, for the Sender, her current type. A Markov strategy for the Sender is a collection of maps $\sigma_i^S : [0,1] \to \Delta(M)$, and a Markov strategy for the Receiver is a collection of maps $\sigma_i^R : M \to A$, one for each $i \in \Omega$. The Receiver's belief about x_t after message m in state i is $\mu_i(\cdot \mid m) \in \Delta([0,1])$. Because the environment is stationary and strategies are Markov, these maps do not depend on calendar time.

A profile of Markov strategies (σ^S, σ^R) and beliefs μ form an MPE if the following conditions hold:

(i) *Sender optimality*: given σ^R, in every period σ^S maximises the Sender's payoff in equation (2.2), measured from that period onwards;

(ii) *Receiver optimality*: for every $i \in \Omega$ and $m \in M$, $\sigma_i^R(m) \in \arg\max_{a \in A} \int u(a,x)\, \mathrm{d}\mu_i(x \mid m)$, with ties broken in favour of $a = 0$;

(iii) *Consistency*: for every $i \in \Omega$, $\mu_i(\cdot \mid m)$ is derived from the state-i type distribution and σ_i^S by Bayes' rule whenever m is sent with positive probability. No restriction is placed on beliefs after messages sent with zero probability.

Three remarks complete the description. First, since the Receiver has only two actions, restricting to two messages is without loss of generality for equilibrium outcomes: messages that induce the same action can be merged. I therefore take $M = \{\ell, h\}$, where ℓ recommends the low action and h recommends the high action, and I use "message" and "recommendation" interchangeably.

Second, as in any cheap-talk game, a babbling MPE always exists. In it, the Sender's messages are uninformative, and the Receiver takes his prior-optimal action: the low action in state L and the high action in state H, by Assumption 2. The question of interest is therefore when informative MPE exist, and what they look like.

Third, the restriction to Markov strategies is standard in stochastic games, and it makes communication depend only on the payoff-relevant state. It is also natural here: Receivers never observe the Sender's past types, so they cannot check her past recommendations, which limits the scope for history-dependent rewards and punishments. Whether non-Markov equilibria can support more communication is left for future work.

Discussion and an application. The model separates the variable that is communicated about, the Sender's type, from the variable that evolves endogenously, the distribution of types. The state is publicly observed and summarises all of the Receiver's external information, so it pins down his prior. Through her messages, the Sender influences the Receiver's action, and hence the distribution from which future types are drawn. Because the Sender takes this into account and the Receiver does not, the two may disagree about which action should be taken, even though their stage payoffs coincide. The Sender then faces a trade-off: misreporting lowers today's payoff but can raise expected future payoffs by shifting the state.

As a running example, consider a government (the Sender) advising a population of citizens (the Receivers) during an epidemic. The state describes the phase of the epidemic: it is either contained ($\omega_t = H$) or surging ($\omega_t = L$). The phase is publicly visible, for instance through hospital admissions or reported pressure on health services. The type x_t is the share of the population that is currently healthy. If citizens knew it, they would be better off when the epidemic is contained than when it is surging (Assumption 1). Through its testing programme, the government observes x_t; citizens observe only the phase. Each period, citizens decide whether to stay at home ($a_t = 0$) or go out ($a_t = 1$), and going out is worthwhile when enough of the population is healthy. Going out also raises transmission, and hence the probability that the epidemic is surging next period: $\varepsilon_L, \varepsilon_H < 0$.[6] Each citizen is negligible, so none takes this effect into account; the population therefore behaves like the model's short-lived Receiver. A benevolent government maximises the citizens' discounted payoffs, which is the Sender's objective in equation (2.2).

[6] Positive association arises, for instance, when a mentor advises a sequence of project-specific contractors on whether to work hard on a project. Hard work on good projects builds the team's reputation and attracts better projects in future ($\varepsilon_i > 0$), and the mentor cares about the contractors' payoffs.

3 A Two-period Model

To build intuition, I first study a version of the model in which the game ends after two periods. To keep the algebra simple, there is no discounting: in place of equation (2.2), the Sender's payoff is the sum of the two Receivers' payoffs. Strategies may now depend on the period as well as on the state, and an equilibrium is defined by conditions (i)–(iii) of Section 2, applied period by period.

By equation (2.1), in any period and state i, the Receiver's best response to a message m is:

$$a = 1 \quad \text{if and only if} \quad \mathbb{E}_i[x \mid m] > \frac{1}{2}, \tag{3.1}$$

where the expectation is taken with respect to his belief $\mu_i(\cdot \mid m)$. In particular, the only information that matters to the Receiver is whether the Sender's type is above or below $1/2$.

I solve the game backwards. In the second period there is no future, so the Sender has no reason to mislead the Receiver, and she reveals what he needs to know (Section 3.1). In the first period, the Sender also cares about the second-period state, which the Receiver ignores. As a result, she prefers a different action from the one the Receiver would choose if he knew her type. Since the Receiver anticipates this, the Sender's recommendations are credible only if her preferred action is not too different from his (Section 3.2). Section 3.3 illustrates these results.

3.1 The second period

In the second period the game ends after the Receiver's action, so the Sender's and the Receiver's payoffs coincide. It is therefore an equilibrium for the Sender to recommend the high action if and only if her type exceeds $1/2$, and for the Receiver to obey. I refer to this as *full revelation*.[7] Under full revelation, the expected payoff in state i is v_i, defined in equation (2.3). Let $\Delta \coloneqq v_H - v_L$ denote how much better the high state is under full revelation. By Assumption 1, $\Delta > 0$. Moreover, $\Delta < 1$: since $f_L > 0$, $v_L > 0$, and since $2x - 1 < 1$ for $x < 1$, $v_H < 1$.

[7] Full revelation means that the Sender reveals whether her type exceeds $1/2$, which, by condition (3.1), is all the Receiver needs to know. A babbling equilibrium also exists in the second period; I select full revelation, which gives both players their highest payoff in that period.

3.2 The first period

Now consider the first period, in state i. I first ask which action the Sender would like the Receiver to take, and then whether she can credibly communicate it. Since second-period play is fully revealing, the Sender's expected second-period payoff is v_L or v_H, depending on the second-period state. By equation (2.5), the high action raises the probability of the high state by ε_i relative to the low action, and so raises the Sender's expected second-period payoff by $\varepsilon_i\Delta$. A Sender of type x therefore strictly prefers the high action to the low action if and only if $2x - 1 + \varepsilon_i\Delta > 0$, that is, if and only if $x > \hat{\theta}_i$, where

$$\hat{\theta}_i \coloneqq \frac{1}{2} - \frac{\varepsilon_i\Delta}{2}. \tag{3.2}$$

The threshold $\hat{\theta}_i$ is the cutoff the Sender would use if she could choose the first-period action herself. It differs from the Receiver's cutoff, $1/2$, by $-\varepsilon_i\Delta/2$. Since $|\varepsilon_i| \leq 1$ and $\Delta \in (0,1)$, this difference is less than $1/2$ in absolute value, so $\hat{\theta}_i \in (0,1)$: in each state, some types prefer each action. The sign of the difference is the sign of $-\varepsilon_i$: the Sender is biased towards the action that raises the probability of the high state, and the bias is larger the more the action affects the transition ($|\varepsilon_i|$) and the more the high state is worth (Δ). Under positive association, the Sender favours the high action in both states; under negative association, she favours the low action in both states; under mixed association, her bias points in opposite directions in the two states. When transitions are exogenous in state i, which includes the case in which state i is absorbing, $\hat{\theta}_i = 1/2$ and the Sender has no bias in that state.

In the epidemic example, going out raises the probability of a surge ($\varepsilon_L, \varepsilon_H < 0$), so $\hat{\theta}_L, \hat{\theta}_H > 1/2$. The government recommends staying at home more often than it would in a static setting: in effect, it overstates the share of the population that is infected.

The Sender would therefore like to tell the Receiver whether $x > \hat{\theta}_i$, but she cannot commit to doing so. Whether such a recommendation is credible depends on whether the Receiver is willing to follow it. Suppose that she sends ℓ if $x \leq \hat{\theta}_i$ and h if $x > \hat{\theta}_i$. By condition (3.1), the Receiver obeys both recommendations if and only if

$$\mathbb{E}_i[x \mid x \leq \hat{\theta}_i] \leq \frac{1}{2} \qquad \text{and} \qquad \mathbb{E}_i[x \mid x > \hat{\theta}_i] > \frac{1}{2}. \tag{3.3}$$

I call these the obedience constraints for ℓ and for h, respectively. Truncating a distribution from above lowers its mean, and truncating it from below raises it. Hence, by Assumption 2, $\mathbb{E}_L[x \mid x \leq \hat{\theta}_L] \leq q_L \leq 1/2$ and $\mathbb{E}_H[x \mid x > \hat{\theta}_H] \geq q_H > 1/2$: only the obedience constraint for h can bind in state L, and only the obedience constraint for ℓ can bind in state H. Define the *critical values* c_L and c_H by:

$$\mathbb{E}_L[x \mid x > c_L] = \frac{1}{2}, \qquad \mathbb{E}_H[x \mid x \leq c_H] = \frac{1}{2}. \tag{3.4}$$

Lemma 1. The critical values c_L and c_H are uniquely defined, with $c_L \in [0, 1/2)$ and $c_H \in (1/2, 1)$. Both obedience constraints hold in state L if and only if $\hat{\theta}_L > c_L$, and in state H if and only if $\hat{\theta}_H \leq c_H$.

Thus a recommendation of the high action is credible in state L if and only if the cutoff exceeds c_L, and a recommendation of the low action is credible in state H if and only if the cutoff is at most c_H.

Lemma 1 gives the conditions under which the Sender's preferred recommendation is credible. The next result shows that these are also the conditions under which an informative equilibrium exists in the first period: when they fail, the Receiver takes his prior-optimal action whatever the Sender says.

Proposition 1. In the first period:

(a) In state L, if $\hat{\theta}_L > c_L$, there is an informative equilibrium, in which the Sender recommends the high action if and only if $x > \hat{\theta}_L$, and the Receiver obeys. If $\hat{\theta}_L \leq c_L$, the Receiver takes the low action in every equilibrium.

(b) In state H, if $\hat{\theta}_H \leq c_H$, there is an informative equilibrium, in which the Sender recommends the high action if and only if $x > \hat{\theta}_H$, and the Receiver obeys. If $\hat{\theta}_H > c_H$, the Receiver takes the high action in every equilibrium.

The logic is the following. In any equilibrium in which the Sender's messages induce both actions, every type above $\hat{\theta}_i$ strictly prefers the high action and every type below $\hat{\theta}_i$ strictly prefers the low action, so the Sender's messages must split types at $\hat{\theta}_i$. Such an equilibrium exists if

and only if the cutoff $\hat{\theta}_i$ satisfies the obedience constraints. The Sender cannot gain credibility by announcing a different cutoff: given the Receiver's response, she would deviate from it. When the constraint fails, no message can move the Receiver away from his prior-optimal action.

Combining Proposition 1 with equation (3.2) gives conditions on the primitives.

Corollary 1. In the first period, an informative equilibrium exists in state L if and only if $\varepsilon_L < (1-2c_L)/\Delta$, and in state H if and only if $\varepsilon_H \geq -(2c_H-1)/\Delta$. In particular, it exists in state L whenever $\varepsilon_L \leq 0$, and in state H whenever $\varepsilon_H \geq 0$.

Proof. By equation (3.2) and $\Delta > 0$, $\hat{\theta}_L > c_L$ if and only if $\varepsilon_L < (1-2c_L)/\Delta$, and $\hat{\theta}_H \leq c_H$ if and only if $\varepsilon_H \geq -(2c_H-1)/\Delta$. Since $c_L < 1/2 < c_H$ by Lemma 1, the first bound is positive and the second is negative. ■

Communication therefore breaks down only when the Sender's bias works against the Receiver's prior: towards the high action in state L, where the Receiver is inclined to take the low action, and towards the low action in state H, where he is inclined to take the high action. A bias in the same direction as the Receiver's prior never undermines communication.

Finally, say that the Sender attains her *first-best* payoff if her payoff equals the one she would obtain by choosing both actions herself.

Corollary 2. Starting from state i, the Sender attains her first-best payoff if and only if there is informative communication in state i in the first period.

3.3 Examples

The following examples show that whether communication survives depends on the type distributions as well as on the transition probabilities.

Example 1. Let $F_L(x) = x$ and $F_H(x) = x^2$, so that $q_L = 1/2$ and $q_H = 2/3$. Then $v_L = 1/4$, $v_H = 5/12$ and $\Delta = 1/6$, so $\hat{\theta}_i = 1/2 - \varepsilon_i/12 \in [5/12, 7/12]$. The critical values are $c_L = 0$ and $c_H = 3/4$. By Corollary 1, informative communication requires $\varepsilon_L < 6$ and $\varepsilon_H \geq -3$, which hold for all transition probabilities: there is informative communication in both states, and the Sender attains her first-best payoff.

Example 2. Let $F_L(x) = x$ and $F_H(x) = x^4$, so that $q_L = 1/2$ and $q_H = 4/5$. Then $v_L = 1/4$, $v_H = 49/80$ and $\Delta = 29/80$, so $\hat{\theta}_i = 1/2 - 29\varepsilon_i/160$. The critical values are $c_L = 0$ and $c_H = 5/8$. Communication in state L is always possible, since $\varepsilon_L < 80/29$ always holds. In state H, however, informative communication requires $\varepsilon_H \geq -20/29$, that is, $z_H + o_H \geq 9/29$. When the high action makes the low state sufficiently more likely, the Sender's bias towards the low action is so strong that a recommendation of the low action is no longer credible: she babbles, and the Receiver takes the high action. This happens, for instance, at the extreme $z_H = o_H = 0$, so that $\varepsilon_H = -1$: the high action leads to the low state, and the low action leads to the high state, with probability one.

In terms of the epidemic example, consider a contained epidemic in which going out is very likely to trigger a surge. The government then wants citizens to stay at home even when most of the population is healthy. Citizens understand this. If the government recommended staying at home whenever it wanted them to, it would do so in many periods in which going out is in fact worthwhile, and on average going out would remain worthwhile even after such a recommendation. A recommendation to stay at home is therefore not credible, and citizens go out whatever the government recommends, which is exactly the outcome the government wanted to prevent.

4 The Infinite-horizon Model

The two-period model captures the Sender's trade-off, but its equilibrium is not stationary: unless $\varepsilon_i = 0$, the Sender uses the cutoff $\hat{\theta}_i$ in the first period and $1/2$ in the second. Moreover, her continuation values are pinned down by full revelation in the second period, whereas in a longer game future communication is itself distorted, and the effect of today's action on the state persists over many periods. I now return to the infinite-horizon model of Section 2, in which the Sender's cutoffs and continuation values are determined jointly and are the same in every period.

Section 4.1 shows that every MPE is described by a pair of cutoffs, one for each state, and expresses the Sender's payoffs in terms of them. Section 4.2 studies the Sender's first-best, Section 4.3 characterises the MPE, and Section 4.4 revisits Examples 1 and 2.

4.1 Cutoffs and payoffs

Fix an MPE and a state i. Since there are two messages and two actions, the Receiver either takes the same action after both messages, or takes different actions after them. In the first case, the Sender's messages do not affect the action, so the outcome is the same as under babbling. Since the Receiver's action is optimal after each message, it is also optimal under the prior, so, as discussed in Section 2, it is the low action in state L and the high action in state H. In the second case, labelling messages by the action they induce, the Receiver follows the Sender's recommendation, and I say that she *communicates* in state i.

In a state in which the Sender communicates, she effectively chooses the action. Let W_i denote her expected payoff in the MPE, defined as in equation (2.2), when the initial state is i. By equation (2.5), inducing the high rather than the low action in state i changes her payoff by:

$$(1-\delta)(2x-1)+\delta\varepsilon_i(W_H-W_L). \tag{4.1}$$

The first term is the Receiver's own gain; the second is the externality, which the Receiver ignores and the Sender internalises. Since equation (4.1) is increasing in x, the Sender induces the high action if and only if $x>\theta_i$, where:[8]

$$\theta_i=\Pi\left(\frac{1}{2}-\frac{\delta\varepsilon_i(W_H-W_L)}{2(1-\delta)}\right), \tag{4.2}$$

and $\Pi(y)\coloneqq\min\{\max\{y,0\},1\}$ is the projection onto $[0,1]$. In a state in which the Sender babbles, the Receiver's action does not depend on her type, which corresponds to $\theta_L=1$ in state L and to $\theta_H=0$ in state H. Either way, the Receiver takes the high action in state i if and only if the Sender's type exceeds θ_i, so every MPE is described by a pair of cutoffs (θ_L,θ_H).

I now express the Sender's payoffs in terms of these cutoffs. Given (θ_L,θ_H), the Receiver's expected payoff in state i is:

$$\tilde{u}_i(\theta_i)\coloneqq\int_{\theta_i}^{1}(2x-1)f_i(x)\,\mathrm{d}x,\qquad i\in\Omega. \tag{4.3}$$

[8] Equation (4.1) compares the two actions in the current period, given that the Sender follows her equilibrium strategy from the next period onwards. By the one-shot deviation principle, which holds in games with discounting and bounded stage payoffs, a strategy that admits no profitable deviation of this kind admits no profitable deviation at all, including deviations to strategies that are not Markov. Condition (i) of the definition of MPE therefore holds if and only if, in every state in which the Sender communicates, she uses the cutoff in equation (4.2).

For instance, $\tilde{u}_L(1/2) = v_L$ and $\tilde{u}_H(1/2) = v_H$ under full revelation, while $\tilde{u}_L(1) = 0$ and $\tilde{u}_H(0) = 2q_H - 1$ under babbling. The probability that the next state is L, given that the current state is i, is:

$$p_i(\theta_i) \coloneqq F_i(\theta_i)z_i + \big(1 - F_i(\theta_i)\big)(1 - o_i), \qquad i \in \Omega. \tag{4.4}$$

Let $\lambda \coloneqq 1 + \delta(p_H(\theta_H) - p_L(\theta_L))$. Since $p_H(\theta_H) - p_L(\theta_L) \geq -1$, $\lambda \geq 1 - \delta > 0$.

Lemma 2. Suppose that, in each state i, the Receiver takes the high action if and only if the Sender's type exceeds θ_i, for some $(\theta_L, \theta_H) \in [0,1]^2$, and let W_i denote the Sender's expected payoff when the initial state is i. Then:

(a) $W_i = \alpha_i \tilde{u}_L(\theta_L) + (1 - \alpha_i)\tilde{u}_H(\theta_H)$ for $i \in \Omega$, where $\alpha_L \coloneqq \big(1 - \delta + \delta p_H(\theta_H)\big)/\lambda$ and $\alpha_H \coloneqq \delta p_H(\theta_H)/\lambda$ satisfy $0 \leq \alpha_H < \alpha_L \leq 1$. In particular:

$$W_H - W_L = \frac{1 - \delta}{\lambda}\big(\tilde{u}_H(\theta_H) - \tilde{u}_L(\theta_L)\big). \tag{4.5}$$

(b) Suppose that $z_L < 1$ and $o_L > 0$, or that $z_H > 0$ and $o_H < 1$. Then the Markov chain over states induced by the cutoffs has a unique invariant distribution, which puts probability

$$\mu \coloneqq \frac{p_H(\theta_H)}{1 + p_H(\theta_H) - p_L(\theta_L)}$$

on state L. Moreover, $\alpha_H \leq \mu \leq \alpha_L$, and $\alpha_L - \alpha_H \to 0$ as $\delta \to 1$, uniformly in (θ_L, θ_H).

Part (a) says that the Sender's payoff is a weighted average of the Receiver's expected payoffs in the two states. The weight on state L is higher when the game starts in L ($\alpha_L > \alpha_H$), because the current period is then spent in state L. Part (b) says that, as the Sender becomes patient, both weights approach the long-run share of time spent in state L. Unlike in models with exogenous states, this share is endogenous: it depends on the cutoffs, and hence on how much the Sender communicates. Finally, equation (4.5) shows that the high state is more valuable to the Sender if and only if it gives the Receiver a higher expected payoff in the current period. The transition probabilities affect only the size of the difference, through λ.

Substituting equation (4.5) into equation (4.2), the Sender's cutoff in a state i in which she

communicates satisfies:

$$\theta_i = \Pi\left(\frac{1}{2} - \frac{\delta\varepsilon_i(\tilde{u}_H(\theta_H) - \tilde{u}_L(\theta_L))}{2\lambda}\right). \tag{4.6}$$

Equation (4.6) is the infinite-horizon counterpart of equation (3.2). In the two-period model, the value of reaching the high state was $\Delta = v_H - v_L$, fixed by full revelation in the second period. Now it is $\tilde{u}_H(\theta_H) - \tilde{u}_L(\theta_L)$, which depends on the cutoffs themselves, so the cutoffs in the two states are determined jointly. It is scaled by δ/λ: since $\sum_{k\geq 0}\delta^k(p_L - p_H)^k = 1/\lambda$, the term $\delta\varepsilon_i/\lambda$ is the effect of the high action today on the expected discounted number of future periods spent in state H. Two questions arise. First, does the system (4.6) have a unique solution? Second, which way does the Sender's bias go? In the two-period model, it pointed towards the high state because $v_H > v_L$. Now the relevant comparison is between $\tilde{u}_H(\theta_H)$ and $\tilde{u}_L(\theta_L)$, and any distortion lowers both.[9]

4.2 The Sender's first-best

As a benchmark, suppose that the Sender could choose the action herself in both states. Her problem is then a discounted dynamic programming problem, and by the argument above, her optimal policy is described by a pair of cutoffs (θ_L^*, θ_H^*) that solve the system (4.6). Let W_L^* and W_H^* denote her first-best payoffs.

Proposition 2. The Sender's first-best has the following properties:

(a) The first-best cutoffs (θ_L^*, θ_H^*) are the unique solution in $[0,1]^2$ to the system (4.6).

(b) $W_H^* > W_L^*$, or equivalently, $\tilde{u}_H(\theta_H^*) > \tilde{u}_L(\theta_L^*)$.

(c) For each $i \in \Omega$, $\theta_i^* < 1/2$ if $\varepsilon_i > 0$, $\theta_i^* = 1/2$ if $\varepsilon_i = 0$, and $\theta_i^* > 1/2$ if $\varepsilon_i < 0$.

Part (a) answers the first question. Although the system (4.6) is nonlinear, it has only one solution. The reason is that any solution describes a policy that is optimal given the continuation

[9] Assumption 1 does not order $\tilde{u}_H(\theta_H)$ and $\tilde{u}_L(\theta_L)$ for arbitrary cutoffs: in Example 1, for instance, $\tilde{u}_H(\theta_H) < v_L = \tilde{u}_L(1/2)$ whenever $\theta_H > 0.84$.

values that it generates itself. Such a policy satisfies the Bellman equation, which has a unique solution (see Appendix B).

Parts (b) and (c) answer the second question: whatever the transition probabilities, the Sender is biased towards the action that raises the probability of the high state, as in the two-period model. Part (b) does not follow from Assumption 1 for arbitrary cutoffs, as footnote 9 shows. Instead, it follows from optimality.

Proof of parts (b) and (c). Two observations are used. First, in any period, the Receiver's expected payoff in state i is at most v_i, which he obtains by taking the high action if and only if $x > 1/2$. Second, whatever the action, the Sender's continuation payoff from the next period onwards is a weighted average of W_L^* and W_H^*, so it lies between them.

Suppose, to the contrary, that $W_L^* \geq W_H^*$, so that state L is at least as valuable as state H. In state L, the Sender's payoff in the current period is at most v_L, and her continuation payoff is at most $W_L^* = \max\{W_L^*, W_H^*\}$. Hence $W_L^* \leq (1-\delta)v_L + \delta W_L^*$, that is, $W_L^* \leq v_L$. In state H, the Sender can induce the high action when $x > 1/2$ in the current period, which gives v_H, and behave optimally from the next period onwards, which gives at least $W_H^* = \min\{W_L^*, W_H^*\}$. Since W_H^* is her optimal payoff, $W_H^* \geq (1-\delta)v_H + \delta W_H^*$, that is, $W_H^* \geq v_H$. By Assumption 1, $W_H^* \geq v_H > v_L \geq W_L^*$, which contradicts $W_L^* \geq W_H^*$.

The equivalence in part (b) follows from equation (4.5), and part (c) follows from part (b) and equation (4.2). ■

Thus, at the first-best, Assumption 1 is enough to determine the direction of the bias. Unlike in the two-period model, however, the size of the bias is not bounded by 1/2, and the first-best cutoffs need not be interior: when the Sender is patient, she may prefer the same action for every type.[10]

[10]For example, let $F_L(x) = 0.9\big(1-(1-x)^9\big) + 0.1x$ and $F_H(x) = x^4$, which satisfy Assumptions 1 and 2. Let $z_L = o_L = 1$, so that in state L the high action leads to state H and the low action keeps the state at L, and let state H be absorbing ($z_H = 0$ and $o_H = 1$). For $\delta = 0.9$, the first-best cutoffs are $\theta_L^* = 0$ and $\theta_H^* = 1/2$: in state L, the Sender wants the high action for every type.

4.3 Equilibrium

I now return to the game. Since the Sender either communicates or babbles in each state, there are four types of MPE: she communicates in both states, only in state L, only in state H, or in neither, which is the babbling MPE. In a state in which she communicates, the Receiver follows her recommendation, so she effectively chooses the action; in a state in which she babbles, the action is fixed at the babbling action. In each type of MPE, the Sender's problem is therefore the problem of Section 4.2, except that she chooses the action only in the states in which she communicates. By the argument of Proposition 2(a), this problem has a unique solution. If she communicates in both states, her cutoffs are the first-best cutoffs (θ_L^*, θ_H^*). If she communicates only in state L, let θ_L' denote her cutoff in state L, given by the system (4.6) with $\theta_H = 0$. If she communicates only in state H, let θ_H' denote her cutoff in state H, given by the system (4.6) with $\theta_L = 1$.

Proposition 3. The MPE of the infinite-horizon model are characterised as follows:

(a) There is an MPE in which the Sender communicates in both states if and only if $\theta_L^* > c_L$ and $\theta_H^* \leq c_H$; only in state L if and only if $\theta_L' > c_L$; and only in state H if and only if $\theta_H' \leq c_H$. The babbling MPE always exists. In each type of MPE, the Sender's cutoffs are those given above, so there are at most four MPE outcomes.

(b) In every MPE in which the Sender communicates in state H, and in the babbling MPE, $W_H > W_L$. In the MPE in which she communicates only in state L, $W_H > W_L$ if and only if $\tilde{u}_L(\theta_L') < 2q_H - 1$, which holds whenever $v_L < 2q_H - 1$.

Part (a) extends Proposition 1: in each state, communication is possible if and only if the Sender's preferred cutoff passes the Receiver's obedience test.[11] Equivalently, the Receiver must be better off following the Sender's recommendations than ignoring them: $\tilde{u}_L(\theta_L) > \tilde{u}_L(1)$ if $\theta_L < 1$ and $\tilde{u}_H(\theta_H) \geq \tilde{u}_H(0)$. Unlike in the two-period model, however, the Sender's preferred cutoff in one state depends on whether she communicates in the other, since communication in a state changes

[11] As in Lemma 1, the obedience constraints hold in state L if and only if $\theta_L > c_L$, and in state H if and only if $\theta_H \leq c_H$. This remains true for cutoffs at the boundary of $[0, 1]$, since a message sent with probability zero imposes no obedience constraint. When the Sender's cutoff is 1 in state L or 0 in state H, communication in that state is outcome-equivalent to babbling: the Sender sends the same message for every type.

the value of being in it. The existence conditions are therefore joint across states. Moreover, several MPE can coexist, and the babbling MPE is always one of them.

Part (b) extends Proposition 2(b).

Proof of part (b). If the Sender communicates in both states, her cutoffs are the first-best cutoffs, so $W_H > W_L$ by Proposition 2(b). If she babbles in both states, $\tilde{u}_H(0) - \tilde{u}_L(1) = 2q_H - 1 > 0$ by Assumption 2, so $W_H > W_L$ by equation (4.5). If she babbles only in state L, the Receiver's expected payoff in state L is $\tilde{u}_L(1) = 0$ in every period, while $\tilde{u}_H(\theta'_H) > 0$, so $W_H > W_L$ by equation (4.5). To see that $\tilde{u}_H(\theta'_H) > 0$, note that $\tilde{u}_H(\theta) > 0$ for every $\theta < 1$: for $\theta \geq 1/2$, the integrand in equation (4.3) is positive on $(\theta, 1)$, and for $\theta < 1/2$, $\tilde{u}_H(\theta) \geq \tilde{u}_H(0) = 2q_H - 1 > 0$, since $\tilde{u}_H$ is increasing on $[0, 1/2]$. Moreover, $\theta'_H < 1$: otherwise $\tilde{u}_H(\theta'_H) = 0$, so $W_H = W_L$ by equation (4.5), and equation (4.2) would give $\theta'_H = 1/2$. Finally, if she babbles only in state H, equation (4.5) gives $W_H > W_L$ if and only if $\tilde{u}_H(0) = 2q_H - 1 > \tilde{u}_L(\theta'_L)$. Since $\tilde{u}_L(\theta) \leq v_L$ for every θ, this holds whenever $v_L < 2q_H - 1$. ■

The exception arises because, when the Sender babbles in state H, the Receiver takes the high action whatever her type, which gives $2q_H - 1 < v_H$. If she communicates in state L, the low state may then become the more valuable one, and her bias in state L reverses. The condition $v_L < 2q_H - 1$ rules this out. It says that the high state is better, even without communication, than the low state with full revelation. It is stronger than Assumption 1, since $v_H \geq 2q_H - 1$, and it holds in both the examples of Section 3.

Proposition 3 also shows when the Sender can attain her first-best payoff.

Corollary 3. The Sender can attain her first-best payoff in an MPE, whatever the initial state, if and only if she can communicate in both states, that is, if and only if $\theta^*_L > c_L$ and $\theta^*_H \leq c_H$.

Corollary 3 is the infinite-horizon counterpart of Corollary 2, with one difference. Since the first-best cutoffs need not be interior, the Sender's preferred action in a state may coincide with the babbling action: in state H if $\theta^*_H = 0$, and in state L if $\theta^*_L = 1$. In these cases, the Sender attains her first-best payoff even though her messages in that state carry no information.

Finally, the Sender reveals exactly what the Receiver needs to know only when her bias vanishes.

Corollary 4. In any MPE in which the Sender communicates in state i and $W_H > W_L$, $\theta_i = 1/2$ if and only if $\varepsilon_i = 0$. In particular, there is an MPE in which $\theta_L = \theta_H = 1/2$ if and only if $\varepsilon_L = \varepsilon_H = 0$.

Proof. The first statement follows from equations (4.5) and (4.6). If $\varepsilon_L = \varepsilon_H = 0$, then $(\theta_L^*, \theta_H^*) = (1/2, 1/2)$ solves the system (4.6), and since $c_L < 1/2 < c_H$, Proposition 3(a) gives an MPE with these cutoffs. Conversely, in an MPE with $\theta_L = \theta_H = 1/2$, the Sender communicates in both states, so $W_H > W_L$ by Proposition 3(b), and the first statement gives $\varepsilon_L = \varepsilon_H = 0$. ■

4.4 Examples

Since the government and the citizens have the same stage-game preferences, any failure of communication is due to the externality. To isolate its role, I revisit Examples 1 and 2, holding the discount factor fixed and varying the size of the externality. In the epidemic application, suppose that staying at home keeps a contained epidemic contained, while going out triggers a surge with probability $s \in (0, 1]$. Suppose also that going out keeps a surging epidemic surging, while staying at home ends the surge with probability $r \in (0, 1]$. That is, $z_H = 0$, $o_H = 1 - s$, $z_L = 1 - r$ and $o_L = 0$, so that $\varepsilon_H = -s$ and $\varepsilon_L = -r$.[12] The parameter s, the risk of a surge, thus measures the externality that citizens impose on the future by going out while the epidemic is contained.

In both examples, $v_L < 2q_H - 1$, so $W_H > W_L$ in every MPE by Proposition 3(b). By equation (4.2), whenever the government communicates, its cutoff exceeds $1/2$: it recommends staying at home more often than citizens would like. In state L, this bias goes in the same direction as the citizens' prior, so communication in state L never fails. The question is whether the government can communicate while the epidemic is contained. In the two-period model, the first-period cutoff in state H is $\hat{\theta}_H = 1/2 + s\Delta/2$, by equation (3.2). In both examples below, $\delta = 0.9$ and $r = 0.3$, and I compute the cutoffs for several values of s.

Example 1 (continued)**.** Recall that $F_L(x) = x$ and $F_H(x) = x^2$, so that $c_H = 3/4$. In the two-period model, $\hat{\theta}_H = 1/2 + s/12 \leq 7/12$, so the government can communicate in state H for every s. Table 1 shows that the same is true in the infinite-horizon model. The government's cutoffs in state H rise with s, but remain below c_H: all four types of MPE exist for every value of s in the table.

[12] Here s is the risk of a *surge* while the epidemic is contained, and r is the probability of *recovery* from a surge.

The cutoffs also differ across MPE. At $s = 0.6$, for instance, the government's cutoff in state H is 0.572 if it also communicates in state L, but 0.654 if it babbles there. When the government cannot communicate during a surge, a surge is more costly, so more is at stake in keeping the epidemic contained, and the government's bias in state H is larger.

s	0.2	0.4	0.6	0.8	1.0
θ_L^*	0.560	0.545	0.536	0.530	0.526
θ_H^*	0.540	0.560	0.572	0.580	0.586
θ_L'	0.527	0.519	0.514	0.512	0.510
θ_H'	0.575	0.622	0.654	0.679	0.699
MPE	all four	all four	all four	all four	all four

Table 1: Cutoffs in Example 1 ($\delta = 0.9$, $r = 0.3$, $c_H = 0.75$).

Example 2 (continued)**.** Recall that $F_L(x) = x$ and $F_H(x) = x^4$, so that $c_H = 5/8$. Table 2 shows that the government's cutoff θ_H^* rises with s, and exceeds c_H when s is above about 0.49. For such values of s, there is no MPE in which the government communicates while the epidemic is contained: citizens go out whatever the government recommends, even though, in every period, they and the government agree on whether going out is worthwhile. Communication in state H breaks down because going out is too likely to trigger a surge, which makes the government's advice to stay at home not credible. In the two-period model, by contrast, $\hat{\theta}_H = 1/2 + 29s/160$, which exceeds c_H only when s is above about 0.69: since the effect of going out persists over many periods, the externality undermines communication at lower values of s in the infinite-horizon model.

s	0.2	0.4	0.6	0.8	1.0
θ_L^*	0.618	0.586	0.567	0.556	0.547
θ_H^*	0.579	0.614	**0.635**	**0.648**	**0.657**
θ_L'	0.610	0.578	0.560	0.549	0.541
θ_H'	0.603	**0.660**	**0.698**	**0.725**	**0.745**
MPE	all four	B, L, LH	B, L	B, L	B, L

Table 2: Cutoffs in Example 2 ($\delta = 0.9$, $r = 0.3$, $c_H = 0.625$). Cutoffs in bold exceed c_H. The last row lists the types of MPE that exist: babbling (B), communication only in state L (L), and communication in both states (LH).

Example 2 also shows why the existence conditions in Proposition 3(a) are joint across states. As in Example 1, the government's cutoff in state H is higher when it babbles in state L than when

it communicates there ($\theta'_H > \theta^*_H$). For s between about 0.26 and 0.49, as at $s = 0.4$ in Table 2, $\theta^*_H \leq c_H < \theta'_H$. There is then an MPE in which the government communicates in both states, but none in which it communicates only in state H: communicating during a surge is what keeps its bias in state H small enough to be credible.

5 Conclusion

This paper studies cheap talk between a long-lived Sender and a sequence of short-lived Receivers when each Receiver's action affects the distribution from which future types are drawn. The Sender and the Receivers have identical stage payoffs, so in a static setting the Sender would tell the Receiver everything he needs to know. With endogenous states, however, the Sender internalises the effect of today's action on the future state, and the Receiver does not. She is therefore biased towards the action that raises the probability of the more valuable state, and her recommendations are credible only if this bias does not work too strongly against the Receiver's prior. In the infinite-horizon model, the Sender's payoffs are weighted averages of the Receiver's expected payoffs in the two states, with weights that converge to an endogenous invariant distribution as she becomes patient. Since the value of each state depends on whether communication is possible in it, the Sender's cutoffs, and hence the existence of informative equilibria, are determined jointly across states. Full revelation is an equilibrium only when actions do not affect transitions. In the epidemic application, a government that cares about future surges recommends staying at home more often than citizens would like, and when going out is too likely to trigger a surge, its advice is ignored altogether.

What drives the results. Three features of the model are essential. First, the Receiver's action must affect transitions: if $\varepsilon_L = \varepsilon_H = 0$, the Sender has no bias. Second, the states must differ in value. If $v_H = v_L$, the argument in the proof of Proposition 2(b) gives $W^*_H = W^*_L$, so the Sender has no bias even when actions affect transitions, and full revelation is her first-best. Third, the Receivers must not internalise the effect of their actions on future states, which is why they are short-lived.

The two assumptions on the type distributions are less restrictive than they may appear. Assumption 1 is without loss of generality, given that the states differ in value: if $v_L > v_H$, the same argument gives $W_L^* > W_H^*$, and the Sender is biased towards the action that raises the probability of state L. Assumption 2 selects the case in which the uninformed Receiver takes different actions in the two states, which fixes the obedience constraint that can bind in each. If instead $q_L, q_H > 1/2$, the uninformed Receiver takes the high action in both states, only the obedience constraint for ℓ can bind, and communication fails in a state only when the Sender's bias towards the low action is too strong there. In either case, as in Corollary 1, communication breaks down only when the Sender's bias works against the Receiver's prior. What changes is the ranking of the states under babbling, which enters Proposition 3(b).

The remaining assumptions are made for tractability. The quadratic stage payoff matters only through the difference $u(1, x) - u(0, x) = 2x - 1$: any stage payoff for which this difference is increasing in x and changes sign in $(0, 1)$ gives the same results, with conditional expectations of the payoff difference in place of conditional means. Continuous type distributions with positive densities ensure that the critical values are unique and that the Sender is indifferent between the two actions with probability zero.

More than two states. With finitely many states, the structure of the analysis carries over. In each state, the Sender uses a cutoff, and an MPE is described by the set of states in which she communicates. The cutoffs are unique for each such set, and the Sender's payoffs are weighted averages of the Receiver's expected payoffs in all states, with weights that converge to the invariant distribution as $\delta \to 1$. What changes is the direction of the bias. With two states, every continuation payoff lies between W_L^* and W_H^*, so the state that is better under full information is also more valuable to the Sender at her first-best. With more states, the value of a state also depends on where it leads: a state that is good under full information but is followed by a bad one can be worth less than a mediocre state that persists. The direction of the bias, and hence which recommendations are credible, then depends on the transitions as well as on the type distributions. For the same reason, full revelation requires only that, in every state, the action does not affect the expected value of the next state, rather than that it leaves the transitions unchanged.

More than two actions. With binary actions, communication in a state is all or nothing: either the Receiver follows the Sender's recommendation, and she obtains her preferred action for every type, or her messages are ignored. This is what keeps the existence conditions tractable, and what allows the Sender to attain her first-best in some equilibria. With richer action spaces, the externality would act as an endogenous counterpart of the Sender's bias in Crawford and Sobel (1982), and communication would typically be partial. The forces identified here would remain: the Sender would be biased towards the more valuable state, and the size of her bias in each state would depend on how much information is transmitted in the others.

Other directions. Several other extensions are natural. First, the state is publicly observed, which keeps the analysis stationary and pins down each Receiver's prior. If only the Sender observed it, her messages would also convey information about the state, and the Receivers' beliefs about it would evolve with the history of messages. Second, I restrict attention to Markov equilibria, and whether history-dependent strategies can sustain more communication remains open. Third, transitions depend only on the current action. If, instead, a surge were triggered only after several consecutive periods of activity, the relevant history could be summarised by a larger set of states, and the issues raised above for more than two states would apply. Finally, the Sender and the Receivers have identical stage payoffs. With a conflict of interest in the stage game, the externality would add to the static bias, and could either offset it, making the Sender more credible, or reinforce it.

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Appendix A Proofs of Section 3

Proof of Lemma 1. The map $c \mapsto \mathbb{E}_L[x \mid x > c]$ is continuous and strictly increasing on $[0,1)$, since its derivative is $\frac{f_L(c)}{1-F_L(c)}(\mathbb{E}_L[x \mid x > c] - c) > 0$. It equals $q_L \leq 1/2$ at $c = 0$, and it exceeds $1/2$ at $c = 1/2$. Hence c_L exists, is unique, and lies in $[0, 1/2)$. Symmetrically, $c \mapsto \mathbb{E}_H[x \mid x \leq c]$ is continuous and strictly increasing on $(0,1]$, is below $1/2$ at $c = 1/2$, and equals $q_H > 1/2$ at $c = 1$, so c_H exists, is unique, and lies in $(1/2, 1)$.

For the obedience constraints, fix a state i, and recall from Section 3.2 that $\hat{\theta}_i \in (0,1)$. Since $\mathbb{E}_i[x \mid x \leq \hat{\theta}_i] \leq \hat{\theta}_i \leq \mathbb{E}_i[x \mid x > \hat{\theta}_i]$ and q_i is a weighted average of the two conditional means, $\mathbb{E}_i[x \mid x \leq \hat{\theta}_i] \leq q_i \leq \mathbb{E}_i[x \mid x > \hat{\theta}_i]$. In state L, the obedience constraint for ℓ therefore holds by Assumption 2, and by strict monotonicity, the obedience constraint for h holds if and only if $\hat{\theta}_L > c_L$. In state H, the obedience constraint for h holds by Assumption 2, and the obedience constraint for ℓ holds if and only if $\hat{\theta}_H \leq c_H$. ■

Proof of Proposition 1. I prove part (a); the proof of part (b) is symmetric, with the obedience constraint for ℓ in place of the one for h.

Existence. Suppose $\hat{\theta}_L > c_L$. Let the Sender send ℓ if $x \leq \hat{\theta}_L$ and h otherwise, and let the Receiver take action 0 after ℓ and action 1 after h. Since $\hat{\theta}_L \in (0,1)$, both messages are sent with positive probability, so beliefs are given by Bayes' rule, and by Lemma 1, the Receiver's actions are optimal. Given the Receiver's strategy, sending h rather than ℓ changes the Sender's payoff by $2x - 1 + \varepsilon_L \Delta$, so by equation (3.2), her strategy is optimal.

Non-existence. Suppose $\hat{\theta}_L \leq c_L$, and consider any equilibrium. Suppose some message sent with positive probability induces the high action and another induces the low action. Every type $x > \hat{\theta}_L$ strictly prefers the high action and every type $x < \hat{\theta}_L$ strictly prefers the low action, so,

up to a set of types of probability zero, the messages that induce the high action are sent exactly by the types in $(\hat{\theta}_L, 1]$. Each such message m must satisfy $\mathbb{E}_L[x \mid m] > 1/2$ by condition (3.1), and $\mathbb{E}_L[x \mid x > \hat{\theta}_L]$ is a weighted average of these conditional means, so $\mathbb{E}_L[x \mid x > \hat{\theta}_L] > 1/2$. By Lemma 1, this contradicts $\hat{\theta}_L \leq c_L$. Hence all messages sent with positive probability induce the same action. If that action were the high action, every on-path message would satisfy $\mathbb{E}_L[x \mid m] > 1/2$, and averaging over messages would give $q_L > 1/2$, contradicting Assumption 2. So every on-path message induces the low action. Finally, no message sent with probability zero can induce the high action, since otherwise every type above $\hat{\theta}_L$, a set of positive probability, would profitably deviate to it. ■

Proof of Corollary 2. If the Sender chose both actions herself, she would take the high action in the second period if and only if her type exceeds $1/2$, and in the first period if and only if it exceeds $\hat{\theta}_i$. Full revelation implements the former. If communication in state i is informative, Proposition 1 implements the latter, so the Sender attains her first-best payoff. If it is not, the Receiver takes the same action for every first-period type. Since $\hat{\theta}_i \in (0,1)$, there is a set of types of positive probability for whom the Sender strictly prefers the other action, so her payoff is strictly below the first-best. ■

Appendix B Proofs of Section 4

Proof of Lemma 2. Write p_L, p_H, $\tilde{u}_L$ and $\tilde{u}_H$ for their values at (θ_L, θ_H). The Sender's payoffs satisfy:

$$W_L = (1-\delta)\tilde{u}_L + \delta\big(p_L W_L + (1-p_L)W_H\big), \qquad W_H = (1-\delta)\tilde{u}_H + \delta\big(p_H W_L + (1-p_H)W_H\big).$$

Subtracting the first equation from the second gives $(W_H - W_L)\big(1+\delta(p_H - p_L)\big) = (1-\delta)(\tilde{u}_H - \tilde{u}_L)$, which is equation (4.5). Substituting $W_H = W_L + (W_H - W_L)$ into the first equation gives $W_L = \tilde{u}_L + \delta(1-p_L)(\tilde{u}_H - \tilde{u}_L)/\lambda$, so the weight on $\tilde{u}_L$ is $1 - \delta(1-p_L)/\lambda = (1 - \delta + \delta p_H)/\lambda = \alpha_L$. Adding equation (4.5) gives the weight on $\tilde{u}_L$ in W_H as $\alpha_L - (1-\delta)/\lambda = \delta p_H/\lambda = \alpha_H$. Clearly $\alpha_H \geq 0$ and $\alpha_L - \alpha_H = (1-\delta)/\lambda > 0$. Finally, $\alpha_L \leq 1$ if and only if $1 - \delta + \delta p_H \leq 1 + \delta p_H - \delta p_L$, that is,

$p_L \leq 1$.

For part (b), suppose first that $z_L < 1$ and $o_L > 0$. Since p_L is a convex combination of z_L and $1 - o_L$, $p_L \leq \max\{z_L, 1 - o_L\}$, so $1 + p_H - p_L \geq \eta \coloneqq 1 - \max\{z_L, 1 - o_L\} > 0$. If instead $z_H > 0$ and $o_H < 1$, the same bound holds with $\eta \coloneqq \min\{z_H, 1 - o_H\} > 0$, since $p_H \geq \eta$. A probability μ on state L is invariant if and only if $\mu = \mu p_L + (1 - \mu)p_H$, that is, $\mu(1 + p_H - p_L) = p_H$, which has a unique solution because $1 + p_H - p_L > 0$. Direct calculation gives:

$$\alpha_L - \mu = \frac{(1-\delta)(1-p_L)}{\lambda(1+p_H-p_L)} \geq 0, \qquad \mu - \alpha_H = \frac{(1-\delta)p_H}{\lambda(1+p_H-p_L)} \geq 0.$$

Finally, $\lambda \geq 1$ if $p_H \geq p_L$, and $\lambda \geq 1 + p_H - p_L \geq \eta$ otherwise, so $\alpha_L - \alpha_H = (1-\delta)/\lambda \leq (1-\delta)/\eta$, which converges to zero uniformly in (θ_L, θ_H). ■

Proof of Proposition 2(a). For $W = (W_L, W_H) \in \mathbb{R}^2$, define the operator T by:

$$(TW)_i \coloneqq \mathbb{E}_i\Big[\max\big\{\delta\big(z_i W_L + (1-z_i)W_H\big),\ (1-\delta)(2x-1) + \delta\big((1-o_i)W_L + o_i W_H\big)\big\}\Big], \qquad i \in \Omega.$$

By standard results for discounted dynamic programming, T is a contraction with modulus δ in the supremum norm, its unique fixed point W^* is the Sender's first-best payoff, and a policy is optimal if and only if, for almost every type, it chooses an action that attains the maximum in the definition of $(TW^*)_i$.[13] By the argument leading to equation (4.2), the high action is strictly better than the low action if $x > \Pi\big(1/2 - \delta\varepsilon_i(W_H^* - W_L^*)/(2(1-\delta))\big)$, and strictly worse if x is below this value. Since types are drawn from distributions with strictly positive densities, the first-best cutoffs are uniquely determined by W^*, and by Lemma 2, they solve the system (4.6).

Conversely, let $(\theta_L, \theta_H) \in [0,1]^2$ solve the system (4.6), and let W be the payoffs given by Lemma 2. By equation (4.5), $\theta_i = \Pi\big(1/2 - \delta\varepsilon_i(W_H - W_L)/(2(1-\delta))\big)$, so in each state the cutoff policy chooses, for almost every type, an action that attains the maximum in the definition of $(TW)_i$. Since W is the payoff of this policy, W_i is the expectation of the maximised value, that is, $W = TW$. Hence $W = W^*$, and $(\theta_L, \theta_H) = (\theta_L^*, \theta_H^*)$. ■

Proof of Proposition 3(a). Fix one of the four types of MPE, and consider an MPE of that type.

[13] This follows from Blackwell's sufficient conditions (Stokey, Lucas, & Prescott, 1989, Theorem 3.3) and the contraction mapping theorem (Stokey, Lucas, & Prescott, 1989, Theorem 3.2); see Chapter 9 for the stochastic case.

Given the Receiver's strategy, the Sender's problem is a discounted dynamic programming problem in which she chooses the action in the states in which she communicates, and the action is fixed at the babbling action in the other states. By condition (i) of the definition of MPE, her strategy solves this problem. The argument in the proof of Proposition 2(a), with the maximum taken only in the states in which she communicates, shows that the solution is unique and that its cutoffs are those given in Section 4.3. By condition (ii) and the argument of Lemma 1, the Receiver obeys both recommendations in state L if and only if the Sender's cutoff there exceeds c_L, and in state H if and only if her cutoff there is at most c_H. This gives the conditions in part (a).

Conversely, suppose that the relevant conditions hold. In each state in which the Sender communicates, let her send h if and only if her type exceeds her cutoff, and let the Receiver take action 1 after h and action 0 after ℓ. In each state in which she babbles, let her send ℓ for every type, and let the Receiver take the babbling action after both messages. Beliefs after messages sent with positive probability follow from Bayes' rule. After a message sent with probability zero, let the Receiver believe that $x = 0$ if the message is ℓ and $x = 1$ if it is h in a state in which the Sender communicates, and let his belief equal the prior in a state in which she babbles. By the conditions, the argument of Lemma 1 and Assumption 2, the Receiver's strategy is optimal. Given the Receiver's strategy, the Sender's strategy solves her problem, so it satisfies condition (i). Hence the profile is an MPE. Since there are four types of MPE and the cutoffs are unique for each, there are at most four MPE outcomes. ■

Proof of Corollary 3. If $\theta_L^* > c_L$ and $\theta_H^* \leq c_H$, Proposition 3(a) gives an MPE in which the Sender communicates in both states with cutoffs (θ_L^*, θ_H^*), in which the Sender's payoff is her first-best payoff. Conversely, consider an MPE in which the Sender attains her first-best payoff from both initial states. Let $W^* = TW^*$ be as in the proof of Proposition 2(a). Since the Sender's MPE payoffs equal W^*, in each state i, W_i^* is the expectation, over types, of the value of the action chosen by the MPE policy, evaluated with continuation payoffs W^*. This value is at most the maximum in the definition of $(TW^*)_i = W_i^*$, so the MPE policy must attain that maximum, that is, choose the first-best action, for almost every type. In state L, if the Sender communicates, her cutoff is θ_L^*, and by Proposition 3(a), $\theta_L^* > c_L$. If she babbles, the Receiver takes the low action

for every type, so $\theta_L^* = 1 > c_L$. Symmetrically, in state H, either the Sender communicates and $\theta_H^* \leq c_H$, or she babbles and $\theta_H^* = 0 \leq c_H$. ■